

HARI BABU R

AI / ML ENGINEER

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PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on project experience building and deploying end-to-end Machine Learning, Deep Learning, and Generative AI solutions. Skilled in Python, Scikit-learn, TensorFlow, and PyTorch for predictive modeling, and in LangChain, LangGraph, and FAISS for building Retrieval-Augmented Generation (RAG) and LLM-powered applications. Experienced in the full ML lifecycle — data preprocessing, feature engineering, model training, evaluation, and deployment — using MLflow for experiment tracking and Streamlit/FastAPI for production-ready interfaces. Strong foundation in NLP, Agentic AI, and cloud deployment on AWS EC2, with a track record of delivering measurable model performance and building intelligent, user-facing AI systems.

TECHNICAL SKILLS

Programming Languages: Python

Machine Learning: Scikit-learn, Supervised & Unsupervised Learning, Model Evaluation, XGBoost, Feature Engineering

Deep Learning: TensorFlow, PyTorch, ANN, CNN, RNN, LSTM, BiLSTM, GRU, BERT

NLP & Generative AI: Natural Language Processing (NLP), Large Language Models (LLMs), Generative AI, Agentic AI, AI Agents, Hugging Face Transformers, OpenAI API

LLM Frameworks & Tools: LangChain, LangGraph, LangSmith, RAG (Retrieval-Augmented Generation), MLflow

Vector Databases: FAISS

Data Analysis & Databases: NumPy, Pandas, SQL, MongoDB

Data Visualization: Matplotlib, Seaborn, Pyplot

Backend & Deployment: FastAPI, Streamlit, AWS EC2

Web Scraping: Playwright

Version Control: Git, GitHub

PROJECTS

Real Estate Investment Advisor [Link](#)

Key Skills: Python, NumPy, Pandas, EDA, Matplotlib, Seaborn, Scikit-learn, XGBoost, Streamlit, MLflow

- Developed a **Machine Learning** application to classify whether a property is a “Good Investment” and predict future property prices after 5 years.
- Performed data preprocessing, **feature engineering**, **exploratory data analysis (EDA)**, and model evaluation on a real estate dataset.
- Implemented **XGBoost** models achieving an XGBRegressor R^2 score of 0.999985 and XGBClassifier accuracy of 99.90%.
- Integrated **MLflow** for experiment tracking and built an interactive **Streamlit** application for real-time investment recommendations and price forecasting.

Personal AI Knowledge Assistant using RAG [Link](#)

Key Skills: Python, LangChain, LangGraph, FAISS, RAG, OpenAI, Streamlit

- Built a Multi-Document **RAG** (Retrieval-Augmented Generation) chatbot using **LangChain**, **LangGraph**, OpenAI, and **FAISS** to answer user queries from PDF, DOCX, TXT, and web documents with high contextual accuracy.
- Designed an end-to-end retrieval pipeline covering document ingestion, text chunking, **embeddings generation**, vector storage, semantic search, and context-aware response generation.

- Deployed an interactive **Streamlit** application with conversational memory using LangGraph, enabling multi-turn, document-based question answering.

Customer Support Automation using Classification Models and Generative AI [Link](#)

Key Skills: Python, ML, NLP, NLTK, Hugging Face Transformers, OpenAI API, Streamlit, Scikit-learn, TensorFlow/Keras

- Built an AI-powered customer support ticket routing system using **NLP**, TF-IDF, and Machine Learning, classifying tickets into 10 support queues with 67% accuracy using a Random Forest model.
- Implemented and benchmarked multiple ML, **Deep Learning** (RNN, LSTM, BiLSTM, GRU), and **Transformer-based (BERT)** models for multi-class text classification.
- Built an end-to-end **Streamlit** application integrated with **OpenAI GPT** to predict ticket categories and generate automated, professional customer support responses in real time.

Garbage Image Classification Using Deep Learning [Link](#)

Key Skills: Python, CNN, TensorFlow, Keras, Computer Vision, OpenCV, Scikit-learn, MobileNetV2, ResNet50, Transfer Learning

- Developed a **6-class garbage image classification** model using **Deep Learning** and **Computer Vision** for automated waste segregation.
- Applied **image preprocessing**, **data augmentation**, and **transfer learning** (ResNet50 and MobileNetV2) to improve model performance, achieving **81.3% test accuracy** with MobileNetV2.
- Built an interactive **Streamlit** web application that allows users to upload garbage images and receive **real-time waste category predictions**.

CERTIFICATIONS

- IIT-M Artificial Intelligence and Machine Learning Program [Link](#)

EDUCATION

Bachelor of Engineering (B.E.)

2018 – 2022

Hindusthan College of Engineering and Technology, Coimbatore, Tamil Nadu